CASE

Eduardo, 64 · 50-pack-year smoking history · weight loss · right-sided ptosis, miosis, anhidrosis · chest CT: **apical lung mass invading the right superior sulcus · Horner syndrome from Pancoast tumor**

BIO 004 · WEEK 8 · CASE DEEP DIVE

The Autonomic Nervous System

A drooping eyelid, a small pupil, a dry forehead — all on the same side. A lung tumor pressing on the sympathetic chain.

PATIENT CASE

Eduardo is 64, retired welder, **50-pack-year smoker**. He comes in for "**my right eye is half closed**" and unintentional weight loss over the past few months. His wife noticed he wasn't sweating on the right side of his forehead.

Exam reveals **right-sided ptosis, miosis (small pupil), and anhidrosis (loss of sweating)** on the right face and forehead — the classic triad of **Horner syndrome**. Chest CT shows a **4-cm apical lung mass invading the right superior pulmonary sulcus**. The diagnosis is **Pancoast tumor with Horner syndrome from invasion of the sympathetic chain at T1**.

Eduardo's findings are the autonomic nervous system showing a single side of one branch (sympathetic) being interrupted. To understand the syndrome you have to know the two limbs of the ANS, where each one originates, where each one synapses, and which targets each one reaches.

> **GUIDING QUESTION**
>
> What are the two branches of the autonomic nervous system, where do they originate, what neurotransmitters do they use, and how does loss of sympathetic outflow to one side of the head produce the Horner triad?

PREWORK Companion to your Week 8 Autonomic Nervous System prework page.

1 TWO LIMBS, OPPOSITE EFFECTS

Sympathetic and parasympathetic

The ANS controls involuntary functions (heart rate, blood pressure, digestion, sweating, pupil size, sexual function). It has two branches that often (but not always) work in opposition:

Sympathetic ("fight or flight")
Mobilizes energy. Speeds heart, dilates pupils, dilates bronchi, slows digestion, sweats, raises BP.

Parasympathetic ("rest and digest")
Conserves energy. Slows heart, constricts pupils, constricts bronchi, stimulates digestion.

Most viscera receive both inputs (dual innervation), and balance between them sets the resting tone.

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2 TWO-NEURON CHAIN

Preganglionic + postganglionic

Both ANS branches use a two-neuron chain to reach the target organ:

- **Preganglionic neuron**: cell body in CNS; axon exits to a ganglion outside CNS. Releases ACh.
- **Postganglionic neuron**: cell body in ganglion; axon goes to the target organ. Releases ACh (parasympathetic) or NE (sympathetic; some exceptions including sweat glands which use ACh).

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3 **SYMPATHETIC**

Thoracolumbar outflow, paravertebral chain

Origin

Lateral horn of the spinal cord

T1-L2

. "Thoracolumbar."

Sympathetic chain (paravertebral ganglia)

Runs alongside the entire vertebral column on both sides. Preganglionic fibers can synapse at the level of entry, ascend, descend, or pass through to splanchnic nerves and synapse in prevertebral ganglia.

Cervical sympathetic ganglia

Superior, middle, inferior (often fused with T1 as the *stellate ganglion*). Fibers from T1 enter the chain and ascend to reach the head — this is the path Eduardo's tumor is interrupting.

Adrenal medulla

Special case: preganglionic sympathetic fibers go directly to chromaffin cells of the adrenal medulla, which release epinephrine/NE directly into the bloodstream.

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4 **PARASYMPATHETIC**

Craniosacral outflow

Cranial outflow

Via cranial nerves

III

(pupil constriction, lens accommodation),

VII

(lacrimal, submandibular, sublingual glands),

IX

(parotid),

X (vagus)

— the big one: heart, lungs, GI tract to splenic flexure.

Sacral outflow

S2-S4 spinal nerves to descending colon, rectum, bladder, reproductive organs.

Parasympathetic ganglia are located IN or NEAR the target organ (long preganglionic, short postganglionic) — opposite of sympathetic (short preganglionic, long postganglionic).

5 NEUROTRANSMITTERS & RECEPTORS

Cholinergic vs adrenergic

SITE	TRANSMITTER	RECEPTOR
All preganglionic (both branches)	ACh	nicotinic
Parasympathetic postganglionic → target	ACh	muscarinic
Sympathetic postganglionic → target (most)	NE	α and β adrenergic
Sympathetic postganglionic → sweat glands	ACh	muscarinic (the cholinergic exception)
Adrenal medulla	Epi/NE into blood	α and β at distant targets

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6 HORNER PATHWAY

Why loss of sympathetic outflow to one eye produces three findings

> **The three findings of Horner syndrome.**
> **Ptosis**: sympathetic innervation supplies the *superior tarsal muscle (Müller's muscle)*, a small smooth muscle that contributes to eyelid elevation. Loss → mild ptosis. (The main eyelid elevator, levator palpebrae superioris, is innervated by CN III, so the ptosis is partial.)
> **Miosis**: sympathetic innervation supplies the *iris dilator muscle*. Loss → unopposed parasympathetic constriction → small pupil.
> **Anhidrosis**: sympathetic cholinergic fibers reach sweat glands of the face. Loss → no sweating on that side of the face.

The sympathetic pathway to the head is a three-neuron chain: hypothalamus → cell body in lateral horn of T1 → preganglionic axon out T1, up the sympathetic chain → superior cervical ganglion → postganglionic axon up the carotid arteries to the orbit. Eduardo's Pancoast tumor compresses the chain at T1, interrupting the second neuron.

✎ **DRAW ZONE 1** -- *Diagram the sympathetic and parasympathetic outflow side by side. Show origin (cord levels), ganglia, and major target organs.*

Sympathetic: T1-L2, short preganglionic, long postganglionic. Parasympathetic: cranial + sacral, long preganglionic, short postganglionic.

DRAW ZONE 2

✎ -- *Trace Eduardo's sympathetic pathway to the right eye: hypothalamus → T1 lateral horn → out T1 root → up sympathetic chain → superior cervical ganglion → up carotid → orbit. Mark where the tumor cuts it.*

Show the three findings of Horner syndrome and which structure each one affects (Müller, iris dilator, facial sweat glands).

RETURNING TO EDUARDO

A lung tumor with a famous neurologic signature

Horner syndrome from a Pancoast (apical) lung tumor is one of the classic clinical presentations in medicine. Every finding maps to interruption of the sympathetic chain.

"*Ptosis*"

Loss of sympathetic input to Müller's muscle. The eyelid no longer gets the boost from this smooth muscle. The lid droops partially (not as severe as CN III palsy ptosis).

"*Miosis*"

Loss of sympathetic input to iris dilator. Parasympathetic constriction (CN III) is now unopposed. Pupil is small. Often more obvious in dim light because the affected pupil can't dilate properly.

"*Anhidrosis*"

Loss of sympathetic cholinergic input to facial sweat glands. The right side of his face and forehead don't sweat.

"*Why Pancoast tumors do this*"

Apical lung tumors invade the superior pulmonary sulcus and adjacent structures — the sympathetic chain (Horner), brachial plexus (T1 = arm weakness/pain), subclavian vessels, ribs.

"*Weight loss + smoking history*"

Suggests malignancy. Apical (superior sulcus) tumors are often non-small-cell lung cancers, frequently adenocarcinoma or squamous cell, often diagnosed late because they're hidden at the apex on chest X-ray.

"*Why it isn't a CN III palsy*"

CN III palsy would produce ptosis + DILATED pupil (not constricted) + eye 'down and out' (not normal position). Horner has the small pupil and normal eye position because the parasympathetic CN III is intact; only sympathetic is out.

Sympathetic, chain, three findings, one tumor. Eduardo's eye is the autonomic nervous system telling you where his cancer is.